

SEQUENCE OF OPERATION (SOO)

INDUSTRIAL BOTTLE PACKAGING LINE AUTOMATION

Document Information

Item	Description
Document Title	Sequence of Operation (SOO)
Project	Industrial Bottle Packaging Line Automation
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1. Introduction

1.1 Purpose

This document describes the complete operational sequence of the Industrial Bottle Packaging Line Automation project.

The objective of this document is to explain how the machine operates during normal production, fault conditions, startup, shutdown, and operator interaction.

The sequence described in this document represents the expected operational behaviour of the machine and should be used as the primary reference during commissioning, testing, troubleshooting, and operator training.

1.2 Scope

This document covers:

- Machine startup
- Automatic production
- Bottle transportation
- Filling operation
- Leak bottle handling
- Capping operation
- Sorting and reject operation
- Alarm response
- Emergency Stop

- Machine shutdown
 - Machine restart
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2. Machine Startup

2.1 Initial Power-Up

After electrical power is applied, the PLC enters the initialization phase.

During initialization:

- The PLC performs its First Scan routine.
- Internal variables are initialized.
- Temporary status bits are cleared.
- Machine outputs remain OFF.
- Production counters retain their previous values unless reset by the operator.
- The machine enters OFF Mode.

The HMI displays that the system is ready for operation.

2.2 Preconditions for Startup

Before automatic production can begin, the following conditions shall be satisfied:

- Emergency Stop is released.
- No active machine faults are present.
- Alarm Reset has been completed if required.
- All actuators are in their home positions.
- The operator has logged into the HMI if configuration changes are required.

If any of these conditions are not satisfied, the machine shall remain in OFF Mode.

2.3 Start Command

The operator initiates production by pressing the Start push button.

The PLC performs the following actions:

- Verifies all startup interlocks.

- Transitions the machine from OFF Mode to Transit Mode.
- Energizes the infeed conveyor.
- Begins automatic bottle transportation.

The production cycle then starts automatically.

3. Automatic Production Sequence

3.1 Bottle Transportation

The conveyor transports bottles through the production line.

The PLC continuously monitors bottle position using the available sensors.

Bottle movement continues until the bottle reaches the filling station.

When the bottle is correctly positioned:

- The conveyor stops.
 - The filling sequence begins.
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3.2 Bottle Position Verification

Before filling starts, the PLC verifies:

- Bottle is present.
- Bottle is correctly positioned.
- No filling-related faults are active.
- The machine remains in Automatic Mode.

Only after successful verification is the filling operation permitted.

3.3 Production Continuity

Following completion of each processing stage, the PLC automatically determines whether the bottle should continue to the next station.

The production cycle continues until:

- The bottle has completed all processing stages.
- A fault interrupts production.

- The operator stops the machine.
- The Emergency Stop is activated.

4. Filling Sequence

4.1 Purpose

The filling sequence is responsible for accurately dispensing the required quantity of liquid into each bottle before it proceeds to the capping station.

The PLC controls the complete filling operation and continuously monitors the filling process to ensure that the preset filling level is achieved.

4.2 Filling Sequence

Step 1 – Bottle Arrival

The bottle is transported by the infeed conveyor toward the filling station.

The PLC continuously monitors the bottle position sensor.

When the bottle reaches the correct filling position:

- The conveyor stops.
 - Bottle movement is inhibited.
 - The filling sequence is initiated.
-

Step 2 – Bottle Verification

Before filling begins, the PLC verifies:

- Bottle is present.
- Machine is operating in Automatic Mode.
- Emergency Stop is not active.
- No active faults prevent filling.

If all conditions are satisfied, the filling operation begins.

Step 3 – Filling Operation

The PLC energizes the filling valve.

Liquid enters the bottle while the filling level is continuously monitored.

The filling operation continues until the preset filling level configured through the HMI has been achieved.

Step 4 – Filling Completion

When the preset level is reached:

- The filling valve is de-energized.
 - Filling stops.
 - The bottle is marked as successfully filled.
 - The PLC prepares the bottle for transportation to the next station.
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Step 5 – Conveyor Restart

The conveyor is restarted.

The bottle leaves the filling station and proceeds toward the capping station.

4.3 Filling Flow Diagram

Bottle Arrives

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Bottle Position Confirmed

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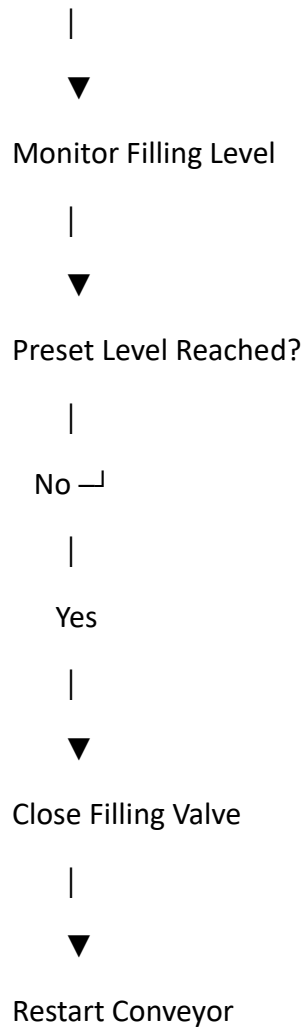


Stop Conveyor

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Open Filling Valve



5. Leak Bottle Handling

5.1 Purpose

The leak bottle function is designed to identify bottles that are considered defective due to liquid leakage.

Unlike immediate machine-stop faults, a leak bottle does not interrupt production.

Instead, the bottle is marked as defective and continues through the remaining production stages before being rejected at the Combined Sorting and Reject Station.

5.2 Leak Detection

Leak bottle status is determined after the filling process has been completed.

When a leak condition is detected:

- The bottle is flagged internally by the PLC.

- Production continues.
- The bottle remains within the production sequence.

5.3 Bottle Tracking

After the bottle has been identified as leaking, the PLC maintains the bottle's defect status while it moves through:

- Conveyor
- Capping Station
- Sorting Station

The bottle is not rejected immediately.

5.4 Reject Decision

When the leaking bottle reaches the Combined Sorting and Reject Station:

The PLC:

- Recognizes the bottle as defective.
- Activates the reject actuator.
- Removes the bottle from the production line.

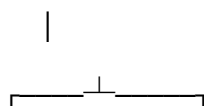
Production continues normally with the following bottles.

5.5 Leak Bottle Flow Diagram

Bottle Filled



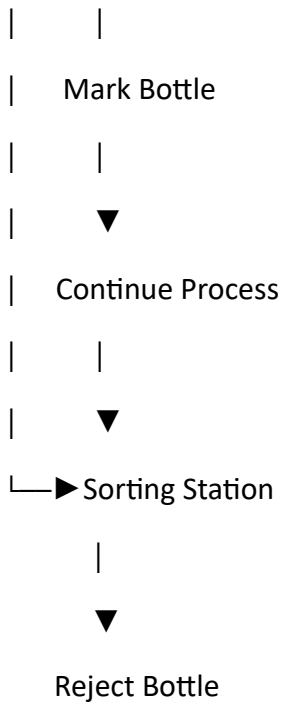
Leak Condition?



No

Yes

No Yes



6. Capping Sequence

6.1 Purpose

The capping sequence automatically applies a cap to each filled bottle before the bottle proceeds to quality verification.

The PLC controls the complete actuator movement and verifies successful completion of the capping cycle.

6.2 Bottle Arrival

After filling has been completed, the conveyor transports the bottle toward the capping station.

When bottle position is confirmed:

- Conveyor stops.
- Capping sequence begins.

6.3 Cap Application

The PLC energizes the capping actuator.

The actuator extends.

A cap is applied to the bottle.

6.4 Actuator Retraction

Following cap application:

The actuator retracts.

The PLC verifies that the actuator has returned to its home position.

Only after successful retraction does the production sequence continue.

6.5 Capping Fault Handling

If the actuator fails to operate correctly:

The PLC generates one of the following faults:

- Capping Arm Retract Jam
- Capping Arm Complete Jam

Machine response depends upon the simulated fault condition.

6.6 Capping Flow Diagram

Bottle Arrives

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Stop Conveyor

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Extend Cap Press

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Apply Cap

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Retract Cap Press



Home Position?



No Yes

Fault Restart Conveyor

7. Sorting and Reject Sequence

7.1 Purpose

The Combined Sorting and Reject Station performs the final product quality evaluation before bottles leave the production line.

7.2 Bottle Arrival

The conveyor transports the bottle to the sorting station.

Bottle position is confirmed before inspection begins.

7.3 Product Evaluation

The PLC evaluates:

- Leak Bottle status.
- Cap Presence status.

These conditions determine whether the bottle is accepted or rejected.

7.4 Accepted Bottle

If the bottle satisfies all quality requirements:

- Reject actuator remains inactive.
 - Conveyor transports the bottle toward the finished product area.
-

7.5 Rejected Bottle

If the bottle is identified as defective:

The PLC:

- Activates the reject actuator.
 - Removes the bottle from the production line.
 - Updates the reject counter.
 - Returns the reject actuator to its home position.
-

7.6 Cycle Completion

After sorting has been completed:

The PLC:

- Updates production statistics.
 - Updates bottles-per-hour.
 - Resets temporary variables.
 - Returns to Transit Mode.
 - Waits for the next bottle.
-

7.7 Sorting Flow Diagram

Bottle Arrives

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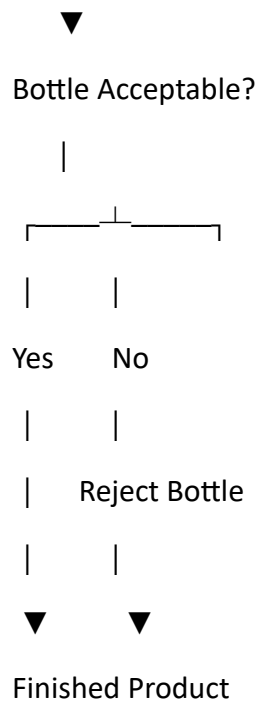


Check Leak Bottle

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Check Cap Present

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8. Alarm Operation

8.1 Purpose

The alarm system continuously monitors the Bottle Packaging Line for abnormal operating conditions that may affect production, machine performance, or product quality. When a fault is detected, the PLC generates the corresponding alarm and responds according to the defined control philosophy.

8.2 Alarm Detection

During every PLC scan, the PLC evaluates machine conditions and compares them with the expected operating state.

When an abnormal condition is detected, the corresponding alarm is generated automatically and displayed on the HMI.

The implemented alarm conditions include:

- Leak Bottle Fault
- Cap Missing Fault
- Capping Arm Retract Jam
- Capping Arm Complete Jam

- Sorting Arm Retract Jam
 - Sorting Arm Complete Jam
 - Emergency Stop
-

8.3 Operator Notification

When an alarm occurs:

- The HMI displays the active alarm message.
- The operator is informed of the fault condition.
- The current machine status is displayed.
- The appropriate machine response is executed.

This allows the operator to quickly identify the cause of the problem and take corrective action.

8.4 Machine Response

The PLC response depends on the type of alarm.

Product Quality Faults

Examples:

- Leak Bottle Fault
- Cap Missing Fault

Response:

- Production continues.
 - The affected bottle is marked as defective.
 - The bottle is rejected at the Combined Sorting and Reject Station.
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Equipment Faults

Examples:

- Capping Arm Retract Jam
- Capping Arm Complete Jam

- Sorting Arm Retract Jam
- Sorting Arm Complete Jam

Response:

- The affected machine operation is stopped.
 - The production sequence is interrupted.
 - The operator must clear the fault before production can resume.
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Emergency Stop

Response:

- All outputs are immediately de-energized.
 - The machine enters a safe stopped condition.
 - Production remains halted until the Emergency Stop is released and the restart procedure is completed.
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8.5 Alarm Recovery

After the fault has been corrected, the operator shall:

1. Verify that the fault has been removed.
2. Release the Emergency Stop (if applicable).
3. Press the Alarm Reset button.
4. Verify that all alarms have been cleared.
5. Press the Start button to resume production.

The machine shall not restart automatically following a fault condition.

In standard industrial practice, machines are not allowed to restart automatically after fault recovery, as this ensures operator safety and prevents unintended operation. However, in this educational setup, to facilitate faster testing and fault analysis, the system is designed such that an alarm reset resumes the process from the point at which it was paused.

9. Emergency Stop Operation

9.1 Purpose

The Emergency Stop provides an immediate method of stopping the Bottle Packaging Line whenever an unsafe condition exists or emergency intervention is required.

9.2 Emergency Stop Activation

When the Emergency Stop push button is pressed:

- The PLC immediately detects the Emergency Stop input.
- The production sequence is interrupted.
- All machine outputs are de-energized.
- Conveyor movement stops.
- Filling operation stops.
- Capping operation stops.
- Sorting operation stops.
- Alarm indication is activated.

The machine remains in a safe stopped condition until operator intervention.

9.3 Emergency Stop Recovery

To restart the machine after an Emergency Stop:

1. Eliminate the unsafe condition.
2. Release the Emergency Stop push button.
3. Verify that the machine is safe to operate.
4. Press the Alarm Reset button.
5. Confirm that no active alarms remain.
6. Press the Start button.

The PLC then resumes normal automatic operation.

Automatic restart is not permitted.

10. Machine Shutdown

10.1 Normal Shutdown

The operator may stop production at any time using the Stop push button.

When the Stop command is received:

- The PLC terminates the production sequence.
- Machine outputs are de-energized in a controlled manner.
- Conveyor motion stops.
- Filling valve closes.
- Capping actuator remains in its home position.
- Reject actuator returns to its home position.
- The machine enters OFF Mode.

The HMI indicates that production has stopped.

10.2 Shutdown Verification

Following shutdown:

- No conveyors are operating.
 - All actuators are in their normal positions.
 - No bottle movement occurs.
 - Machine outputs remain OFF.
 - Production statistics remain available for operator review.
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11. Restart Procedure

11.1 Restart Following Normal Shutdown

To restart production after a normal shutdown:

1. Verify that the machine is ready for operation.
2. Confirm that no active alarms are present.
3. Press the Start button.

The PLC resumes automatic production where it left.

11.2 Restart Following Fault Recovery

If production was interrupted by a fault:

1. Correct the fault condition.
2. Verify safe operating conditions.
3. Perform Alarm Reset.
4. Confirm that all alarms have been cleared.
5. Press the Start button.

The machine resumes automatic production.

12. Operator Responsibilities

The operator is responsible for the safe and efficient operation of the Bottle Packaging Line.

Primary responsibilities include:

- Starting and stopping the machine.
- Monitoring production through the HMI.
- Responding to alarm conditions.
- Resetting alarms after corrective action.
- Configuring production parameters when authorized.
- Monitoring production statistics.
- Performing routine visual inspection of machine operation.
- Ensuring safe operation before restarting production.

The operator shall not attempt to restart the machine until the cause of any fault has been identified and corrected.

13. Sequence Summary

The following table summarizes the complete production sequence.

Step	Operation	PLC Action	Expected Result
1	System Power-Up	Initialize PLC	Machine enters OFF Mode
2	Operator Presses Start	Verify startup conditions	Automatic mode enabled
3	Bottle Transportation	Conveyor operates	Bottle moves to filling station
4	Bottle Positioning	Conveyor stops	Bottle positioned correctly
5	Filling Operation	Filling valve opens	Bottle filled to preset level
6	Leak Evaluation	Bottle status checked	Bottle marked if leaking
7	Bottle Transfer	Conveyor restarts	Bottle moves to capping station
8	Capping Operation	Cap press extends and retracts	Cap applied successfully
9	Sorting Station	Product quality evaluated	Accept or reject decision made
10	Reject Operation	Reject actuator operates if required	Defective bottle removed
11	Production Update	Counters and runtime updated	HMI statistics refreshed
12	Cycle Reset	Temporary variables cleared	Ready for next bottle

14. Conclusion

The Sequence of Operation defines the complete operational behaviour of the Industrial Bottle Packaging Line Automation project from system startup through normal production, fault handling, and machine shutdown.

The document provides a structured description of each stage of the production cycle, including operator interaction, PLC actions, equipment response, and recovery procedures. It serves as a reference for commissioning, testing, troubleshooting, and operator training.

By following the sequence described in this document, the system achieves reliable bottle transportation, filling, capping, quality verification, sorting, production monitoring, and fault management within the intended Proof of Concept environment.

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